

No finite level of the NPA hierarchy is exact for the doubly-tilted CHSH functional near the critical tilt

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Abstract

Gigena, Panwar, Scala, Araújo, Farkas and Chaturvedi [npj Quantum Inf. **11**, 82 (2025)] determined the quantum maximum of the doubly-tilted CHSH functionals by self-testing methods, observed that the Navascués–Pironio–Acín level needed for exactness grows without evident bound toward the critical tilt (level 10 still overshoots at $\alpha = \beta = 0.999$), and asked whether any finite level suffices. We answer this in the negative. For the symmetric critical family $B_s = (1 - \frac{s}{2})(\langle A_0 \rangle + \langle B_0 \rangle) + \text{CHSH}$ we prove: for every NPA level $k \geq 2$ there are an explicit rational $g_k > 0$ and an $s_k^* > 0$ with $c_k(s) \geq 4 - s + g_k s^2$ for all $s \in (0, s_k^*]$; since the quantum value leaves the local bound only cubically, $c_Q(s) = 4 - s + \frac{s^3}{6} + O(s^4)$, every finite level (including 1 and $1+AB$, by monotonicity) strictly overshoots on an interval $(0, \varepsilon_k]$ — no finite level is exact on any neighbourhood of the critical point. For $k \geq 2$ the overshoot exponent is at least 2 unconditionally, and exactly 2 granted the dual second-order expansion of the companion note (the exact almost-quantum coefficient $\frac{3}{64}$); unconditionally $a_2 > \frac{1}{39}$, $a_3 > \frac{1}{188}$, $a_4 > \frac{1}{641}$. The proof is a primal construction: an exactly feasible moment curve $y_0 + s y_1 + s^2 y_2$ at each level, built from level-uniform structural laws and one level-independent *signed* witness — a closed-form class function y^* with $N_k^\top \Gamma(y^*) N_k = u_k u_k^\top$ at every level. The mechanism forces the sign: for $k \geq 3$ no state and no smooth curve of quantum models can realize the gain direction, so the overshoot lives strictly in the non-quantum part of the NPA tangent cone. The proof is computer-assisted in the strict sense: finite exact-integer verifications with proven degree bounds are constituent parts of the argument. Every step is accompanied by a verification program, and the chain has been re-verified against independent implementations, including a symbolic per-regime proof of the witness identity and a clean-room implementation written from this paper’s text alone. The one external input is the published quantum value of Gigena et al., cross-checked to twelve digits with three errata corrected.

1 Introduction

1.1 The question

In the minimal $(2, 2, 2)$ Bell scenario, the doubly-tilted CHSH functionals

$$B_{\alpha\beta} = \alpha \langle A_0 \rangle + \beta \langle B_0 \rangle + \sum_{x,y \in \{0,1\}} (-1)^{xy} \langle A_x B_y \rangle$$

*Exact rational/integer verification code and certificate data: <https://github.com/tohafrit/npa-nonexactness> (see Appendix A).

have local bound $c_L = 2 + \alpha + \beta$ and a quantum advantage exactly for $0 < \alpha + \beta < 2$. Gigena et al. [1] determined the quantum maximum $c_Q(\alpha, \beta)$ in closed form (the largest root of an explicit sextic) together with a self-testing statement, and made a striking numerical observation: approaching the critical line $\alpha + \beta = 2$, the level of the Navascués–Pironio–Acín (NPA) hierarchy [2] required to reach c_Q grows without evident bound — at $\alpha = \beta = 0.999$ even level 10 exceeds c_Q by 8.6×10^{-10} . They left open “if any finite level of NPA will be enough...even in the CHSH scenario.”

This paper proves that no finite level is enough. Throughout we work on the symmetric slice $\alpha = \beta = 1 - s/2$, i.e. with

$$B_s = B_0 - \frac{s}{2}(\langle A_0 \rangle + \langle B_0 \rangle), \quad B_0 = \langle A_0 \rangle + \langle B_0 \rangle + \text{CHSH},$$

$s \in (0, 2)$, $s \rightarrow 0^+$ the critical limit. Let $c_k(s)$ denote the level- k NPA value and $c_Q(s)$ the quantum value. For every level $k \geq 2$, and for the intermediate level $1+AB$, $c_k(0) = 4$ and $c'_k(0^+) = -1$: the critical maximum is attained on a classical face and the dual certificate of Lemma 3.2 pins the value there. (Level 1 is looser, $c_1(0) \approx 4.73$; it enters only through monotonicity at the end.) The quantum curve is *cubically* flat,

$$c_Q(s) = 4 - s + \frac{1}{6}s^3 - \frac{1}{36}s^4 + O(s^5), \quad (1)$$

and $c_k \geq c_Q$ always, so near $s = 0$ every level sits above $4 - s$ with a possible *quadratic* overshoot. Define

$$a_k := \liminf_{s \rightarrow 0^+} \frac{c_k(s) - 4 + s}{s^2} \geq 0.$$

In the companion note [3] we computed the almost-quantum coefficient exactly, $a_{1+AB} = \frac{3}{64}$, certified $a_2 > \frac{1}{39}$, $a_3 > \frac{1}{188}$, $a_4 > \frac{1}{641}$ by exact rational tangent-program certificates (modulo a then-unproven framework step, discharged below), and reduced the open question of [1] to the positivity of the single sequence (a_k) . Here we prove that positivity, for every k .

Theorem 1.1 (Main). *Fix any NPA level $k \geq 2$. There exist explicit rational constants $g_k > 0$ and $s_k^* > 0$ such that*

$$c_k(s) \geq 4 - s + g_k s^2 \quad \text{for all } s \in (0, s_k^*].$$

Consequently $a_k \geq g_k > 0$, and, using the quantum expansion (1) of [1], for every finite level k (including $k = 1$ and $k = 1+AB$) there is an $\varepsilon_k > 0$ with

$$c_k(s) > c_Q(s) \quad \text{for all } s \in (0, \varepsilon_k]:$$

no finite level of the NPA hierarchy is exact on any neighbourhood of the critical point. Moreover, for every $k \geq 2$ the overshoot order is at least two, and exactly two granted the dual second-order expansion $c_{1+AB}(s) = 4 - s + \frac{3}{64}s^2 + o(s^2)$ of the companion note [3]: $g_k s^2 \leq c_k(s) - (4 - s) \leq \frac{3}{64}s^2 + o(s^2)$. Quantitatively, $g_k = 1/(1024 t_0(k))$ for an explicitly computable $t_0(k) < \infty$, and unconditionally

$$a_2 > \frac{1}{39}, \quad a_3 > \frac{1}{188}, \quad a_4 > \frac{1}{641}.$$

1.2 The one external input

The proof of the interval bound $c_k(s) \geq 4 - s + g_k s^2$ is entirely self-contained (finite exact arithmetic plus finite-dimensional convexity). Translating it into *non-exactness* uses one published external input: the closed-form quantum value of Gigena et al. [1], in the form of the

expansion (1). We derived (1) symbolically from their sextic in [3], cross-checked it to twelve digits against their own reported value at $s = 0.002$, and documented (and corrected) three typographical errata in their printed display equations; their characteristic polynomial and all quoted numerics are correct. We use nothing else from outside: in particular no numerical SDP value enters any proof below.

1.3 The proof chain

The argument is layered; every layer is proven for all k and is accompanied by an exact machine verification (Appendix A).

1. **Structural laws** (Section 3). At $s = 0$ the optimal set of the level- k SDP is a k -dimensional face F_k with an explicit affine parametrization (*product law*), an integer relation lattice N_k of dimension $2k^2$ forced in the kernel of every optimal moment matrix (law L1), a distinguished interior base point $y_0(\delta^*)$ whose kernel is *exactly* $\text{span } N_k$ — realized by one explicit commuting-party density operator, uniformly in k (law L2) — and a closed-form rational Slater point for the second-order tangent program (law L4).
2. **The witness** (Section 4). Quadratic gain at level k requires a first-order direction y_1 whose compressed form $N_k^\top \Gamma(y_1) N_k$ is a positive multiple of $u_k u_k^\top$ for a specific level-uniform vector u_k (closed form given). We exhibit one level-*independent* signed class function y^* with $N_k^\top \Gamma(y^*) N_k = u_k u_k^\top$ exactly at *every* level (Theorem 4.4). The proof is by locality lemmas reducing each kernel-pair equation, within finitely many “fine regimes”, to a polynomial of degree ≤ 2 per free length coordinate, plus an exhaustive exact-integer check of a finite interpolation grid — a constituent part of the proof, not an illustration.
3. **Small- λ positivity** (Section 5). Given the witness, an explicit feasible point of the second-order tangent program with positive value $v_k \geq 1/(1024 t_0) > 0$ exists at every level (Theorem 5.4); the block-PSD conditions reduce to Schur complements by a range identity that holds identically on the constraint space.
4. **Exact arc feasibility** (Section 6). A no-repair positivity lemma (Lemma 6.1) shows that the resulting jet $y(s) = y_0 + s y_1 + s^2 y_2$ is *exactly* feasible on an explicit interval $(0, s_k^*]$ — no $o(s^2)$ correction term — with exactly cubic objective $4 - s + g_k s^2 + t_0 \lambda^2 s^3$ (Theorem 6.3). This also discharges, retroactively, the framework caveat of [3], making the ladder bounds $a_2 > \frac{1}{39}$, $a_3 > \frac{1}{188}$, $a_4 > \frac{1}{641}$ unconditional (Corollary 6.4).
5. **Assembly** (Section 7): the chain above yields Theorem 1.1; the sandwich $c_k \leq c_2 \leq c_{1+AB} = 4 - s + \frac{3}{64} s^2 + o(s^2)$ (upper half from the companion’s dual expansion) pins the overshoot exponent to 2.
6. **Mechanism** (Section 8): the witness y^* *must* be signed — no Hilbert-space state and no smooth curve of genuine quantum models can produce the gain direction (in a Krein space the same computation forces the gain-carrying direction to be neutral; see Remark 8.2). The finite-level failure is thus located precisely: finite NPA levels admit a signed tangent direction that quantum mechanics forbids.

The entire chain was subsequently re-checked by an independent verification pass — freshly written code for the witness identity and for a long-length attack on the regime lemmas, independent driver scripts over the shared exact stack for the remaining items (the per-item provenance

is stated in Appendix A); the audit confirmed every step and supplied one further exact check for the single step the original programs did not cover.

1.4 Relation to prior work

That the NPA hierarchy converges *at each fixed* s is not in question (the quantum value is attained in this minimal scenario [5]); our theorem is the divergence of the required level as $s \rightarrow 0^+$, exactly the uniform failure asked about in [1]. The result complements the degree-bounded noncommutative Fejér–Riesz theorem of Klep, Levenson and McCullough [6], whose strict-positivity hypothesis fails precisely in this $s \rightarrow 0$ limit (margin $\sim s^3$); it realizes, at a concrete attained and self-tested optimum, the failure mode left open by Nie’s finite-convergence theorem [7] through Marshall’s boundary Hessian condition [8] — the critical touch is cubic, contact order three, so the generic finite-exactness criterion is silent — and it does so by an unconditional construction rather than a hardness argument [10]. See Section 9.

2 Setup

2.1 Levels, classes, the symmetric picture

The scenario algebra is generated by involutions A_0, A_1 (Alice) and B_0, B_1 (Bob), $X^* = X$, $X^2 = 1$, $[A_i, B_j] = 0$: the group algebra of $\mathbb{Z}_2^{*2} \times \mathbb{Z}_2^{*2}$. A *word pair* is $w = (a, b)$ with a an alternating (= reduced) word in $\{A_0, A_1\}$ and b alternating in $\{B_0, B_1\}$; the *level- k basis* $\mathcal{W}_k$ consists of all word pairs with $|a| + |b| \leq k$. Moments are indexed by *classes* (of total length $\leq 2k$, the products arising in Γ_k): reduced word pairs modulo dagger (reversal of both parts) and party swap. For a moment vector y on classes, the level- k moment matrix is

$$\Gamma_k(y)_{pq} = y[\text{class}(\text{rev}(p) \cdot q)], \quad p, q \in \mathcal{W}_k,$$

with per-party reduction of $\text{rev}(a_p)a_q$ and $\text{rev}(b_p)b_q$. The level- k value is $c_k(s) = \max\{B_s \cdot y : \Gamma_k(y) \succeq 0, y_1 = 1\}$. Working in the party-swap-symmetrized class space is sound and complete for this swap-symmetric functional (average any maximizer with its swap; see the soundness lemma of [4]). In class coordinates,

$$B_0 \cdot y = 2y_{\langle A_0 \rangle} + y_{\langle A_0 B_0 \rangle} + 2y_{\langle A_0 B_1 \rangle} - y_{\langle A_1 B_1 \rangle}, \quad B_s \cdot y = B_0 \cdot y - s m \cdot y,$$

where $m \cdot y := y_{\langle A_0 \rangle}$ (so $s m \cdot y = \frac{s}{2}(\langle A_0 \rangle + \langle B_0 \rangle)$ on swap-symmetric y).

Core coordinates. For an alternating word let $i(\cdot)$ count its 1-letters. Write $\alpha_i = A_1(A_0 A_1)^{i-1}$ and $\beta_j = B_1(B_0 B_1)^{j-1}$ for the 0-free cores, $\delta_j = \langle \beta_j \rangle$ ($\delta_0 := 1$), and $m_{ij} = \langle \alpha_i \beta_j \rangle$.

2.2 The relation lattice

Two families of integer vectors in $\mathbb{R}^{\mathcal{W}_k}$ play the lead role. For a word w ending in 0 (or empty) write w' for w with its trailing 0 removed ($\emptyset' = \emptyset$) and $w.1$ for w with a 1 appended. *Trivial reductions:* for each basis word pair $q = (a, b)$ in which at least one part has a trailing 0, the vector $T_q = e_{(a,b)} - e_{(a^\sharp, b^\sharp)}$, where $\sharp$ removes the trailing 0 from *each* part that has one (and fixes the other). *Dressings:* for each word pair $W = (w_a, w_b)$ in which neither part ends in 1 and $|W| \leq k - 2$, the tensor vector

$$D_W = (e_{w'_a} - e_{w_a.1}) \otimes (e_{w'_b} - e_{w_b.1}) = e_{(w'_a, w'_b)} - e_{(w'_a, w_b.1)} - e_{(w_a.1, w'_b)} + e_{(w_a.1, w_b.1)},$$

which represents $W \cdot (1 - A_1)(1 - B_1) \psi$ in the basis after both the group reduction *and* the state relations $A_0 \psi = B_0 \psi = \psi$ (the trailing-0 strip in the first corner) are applied; the naive group-reduced corners $W \cdot (1, A_1, B_1, A_1 B_1)$ differ from D_W by trivial reductions and would *not* satisfy the exact identities below. Let $N_k = [T\text{'s} \mid D\text{'s}]$ denote the resulting integer matrix, the *relation lattice*.

Proposition 2.1 (counting laws). *For every level $k \geq 2$: $|\mathcal{W}_k| = 2k^2 + 2k + 1$; the number of classes is $\frac{(5k+2)(k+1)}{2}$; there are $\frac{3k^2+k}{2}$ trivial reductions and $\frac{k(k-1)}{2}$ dressings; and the $2k^2$ columns of N_k are linearly independent.*

Proof. Basis: for each pair of lengths (ℓ_a, ℓ_b) with $\ell_a + \ell_b \leq k$ there are two alternating words per positive length and one empty word; summing gives $2k^2 + 2k + 1$. Irreducible words (no trailing 0 in either part): one word per positive length per party ends in 1, giving $1 + 2k$ single-party words and $\frac{k(k-1)}{2}$ two-party words; the trivial-reduction count follows by subtraction. Dressings: the word W must not end in 1 in either part (else the appended R_1 -letter cancels and the vector reproduces a shorter dressing up to sign), so there is one admissible word per length per party with $|W| \leq k - 2$: $\#\{(x, y) \geq 0 : x + y \leq k - 2\} = \frac{k(k-1)}{2}$. Independence: order the basis; each T_q is the unique lattice vector supported on the reducible word q (dressings are supported on irreducible words only, since their corners are written post-reduction), and each dressing has a strictly longest corner (w_a+1, w_b+1) , distinct across dressings. $\square$

3 The structural laws

This section establishes three level-uniform laws about the critical face $s = 0$: the face law (L1), the kernel law (L2), and the uniform Slater law (L4). (The labels match the discovery record and verification suite.)

3.1 L1: the face law

Theorem 3.1 (face law). *For every $k \geq 2$, the set $F_k = \{y : \Gamma_k(y) \succeq 0, y_1 = 1, B_0 \cdot y = 4\}$ satisfies:*

1. $B_0 \cdot y \leq 4$ on $\{\Gamma_k(y) \succeq 0, y_1 = 1\}$, so F_k is the optimal face at $s = 0$;
2. every $y \in F_k$ satisfies $\Gamma_k(y) N_k = 0$;
3. the solution set $\{y : \Gamma_k(y) N_k = 0, y_1 = 1\}$ is a k -dimensional affine family, parametrized by $(\delta_1, \dots, \delta_k)$ via the product law

$$m_{0j} = \delta_j, \quad m_{ij} = \delta_i + \delta_j - 1 \quad (i, j \geq 1),$$

all other moments equal to their core values (0-padding invisible);

4. $B_0 \cdot y = 4$ identically on that affine hull, and the s -linear response $-m \cdot y = -1$ is constant on F_k ; hence F_k is also the $s \rightarrow 0^+$ optimal face, first-order selection is vacuous, and selection starts at order s^2 — the tangent program of Section 5.

The proof occupies Lemmas 3.2–3.6. We write $y_0(\delta)$ for the product-law moment vector with parameters $\delta = (\delta_1, \dots, \delta_k)$.

Lemma 3.2 (rational maximal-rank dual certificate at level $1+AB$). *Let W be the 9×5 integer matrix whose columns represent, in the $1+AB$ basis $\{\psi, A_i\psi, B_j\psi, A_iB_j\psi\}$, the vectors*

$$w_1 = (A_0 - B_0)\psi, \quad w_2 = (A_0B_0 - 1)\psi, \quad w_3 = ((A_0 - 1)B_1 - (B_0 - 1)A_1)\psi,$$

$$w_4 = ((A_0 - 1) + (B_0 - 1) + (A_0 - 1)B_1 + (B_0 - 1)A_1)\psi, \quad w_5 = R_1\psi,$$

and let

$$E = \begin{pmatrix} 13/40 & 0 & 3/40 & 0 & 0 \\ 0 & 1/5 & 0 & 1/20 & 0 \\ 3/40 & 0 & 1/8 & 0 & 0 \\ 0 & 1/20 & 0 & 1/8 & 0 \\ 0 & 0 & 0 & 0 & 1/4 \end{pmatrix}.$$

Then $E \succ 0$ (exact leading minors) and $Z_0^* = WEW^\top$ satisfies the affine identity

$$4y_1 - B_0 \cdot y = \langle Z_0^*, \Gamma_{1+AB}(y) \rangle \quad \text{for all } y$$

(exact class-sum check). Hence $Z_0^* \succeq 0$, $\text{rank } Z_0^* = 5$, and $\text{range } Z_0^* = \text{span } W$.

Remark 3.3. (i) The certificate Z_0 of [3] also has rank 5 but a smaller forced subspace; Z_0^* is built so that its range is exactly the maximal forced subspace. (ii) Rank 6 — range containing $(1 - A_0)\psi$ itself — is impossible: strict complementarity fails at this face (an SDP scan attains maximal minimal eigenvalue 0 with an explicit null direction on the relation span of the one-dimensional face family), which is exactly why the linear-closure step (Lemma 3.4) is needed.

Lemma 3.4 (linear closure at $1+AB$). *If Γ is PSD on the $1+AB$ words with $y_1 = 1$ and $\Gamma W = 0$, then the 17 moment classes are forced to*

$$\langle A_0 \rangle = \langle B_0 \rangle = \langle A_0B_0 \rangle = 1, \quad \langle A_0B_1 \rangle = \langle B_1 \rangle, \quad \langle A_1B_0 \rangle = \langle A_1 \rangle,$$

$$\langle A_0A_1 \rangle = \langle A_1 \rangle, \quad \langle B_0B_1 \rangle = \langle B_1 \rangle, \quad \langle A_1B_1 \rangle = \langle A_0A_1B_0B_1 \rangle = \langle A_1A_0B_0B_1 \rangle = \langle A_1 \rangle + \langle B_1 \rangle - 1,$$

with exactly $(\langle A_1 \rangle, \langle B_1 \rangle)$ free: the linear system $\{\Gamma(y)W = 0, y_1 = 1\}$ has rank 15 of 17 (exact rank check; no positivity is used in this step). In particular $\langle A_0 \rangle = 1$ is forced linearly, although the vector relation $(1 - A_0)\psi = 0$ is not dual-forced (Remark 3.3).

Lemma 3.5 (scalar to vector: 2×2 -minor telescoping). *Let $\Gamma_k \succeq 0$ at any level $k \geq 2$, $y_1 = 1$, and suppose the three scalar identities $\langle A_0 \rangle = 1$, $\langle B_0 \rangle = 1$, $\langle R_1 \rangle := 1 - \langle A_1 \rangle - \langle B_1 \rangle + \langle A_1B_1 \rangle = 0$ hold (Lemma 3.4 applied to the $1+AB$ principal submatrix). Then $\Gamma_k N_k = 0$.*

Proof. Work in the Gram picture: $v_w = \text{Gram vector of basis word } w$. For every basis word w such that wA_0 is in the basis,

$$|v_{wA_0} - v_w|^2 = \Gamma_{wA_0, wA_0} - 2\Gamma_{wA_0, w} + \Gamma_{w, w} = 1 - 2\langle A_0 \rangle + 1 = 0,$$

because $\text{rev}(w) \cdot w$ telescopes to the empty word, so the three entries are $y_1, \langle A_0 \rangle, y_1$ regardless of w . Hence all trailing- A_0 reductions (and trailing- B_0 likewise) lie in $\ker \Gamma_k$. For every dressing word W ,

$$|WR_1\psi|^2 = \sum_{d, d'} s_d s_{d'} \langle (Wd)^\dagger (Wd') \rangle = \langle R_1^\dagger R_1 \rangle = 4\langle R_1 \rangle = 0$$

by the same telescoping ($\text{rev}(W) \cdot W$ cancels), so all dressings lie in $\ker \Gamma_k$. This is part 2 of Theorem 3.1, uniformly in k , from one dual certificate, one linear closure, and 2×2 minors. $\square$

Lemma 3.6 (solution set = product law). *The solution set $\{\Gamma_k(y)N_k = 0, y_1 = 1\}$ (itself an affine subspace) equals the product-law family, of affine dimension exactly k .*

Proof. ($\supseteq$) The product-law vector satisfies $\Gamma(y_0(\delta))N_k = 0$ identically in δ : every scalar consequence of the lattice is either a 0-padding invariance (trivial reductions) or $\langle U_a U_b (1 - A_1)(1 - B_1) \rangle = 0$ with $|U_a| + |U_b| \leq 2k - 2$ (dressings), and the latter expands to the four-point relation $m_{i,j} - m_{i+\varepsilon,j} - m_{i,j+\varepsilon'} + m_{i+\varepsilon,j+\varepsilon'} = 0$ ($\varepsilon, \varepsilon' = \pm 1$ according to the last letters), which $m_{ij} = \delta_i + \delta_j - 1$ satisfies in every case (exact machine check of the identity in δ , $k \leq 4$).

($\subseteq$) The linear system forces the product law: one-sided 0-absorption is available at class level for every class word (split any class word (a, b) , $|a| + |b| \leq 2k$, into two basis halves so that the 0-end to be absorbed lies at the outer end of one half; the relation row (p, T_q) then equates the class to its stripped version), and the dressed relations give the recurrence

$$m_{ij} = m_{i,j-1} + m_{i-1,j} - m_{i-1,j-1}, \quad i, j \geq 1, \quad i + j \leq k + 1$$

(take $U = (\alpha'_i, \beta'_j)$, the cores minus their last letters; length $2(i+j) - 4 \leq 2k - 2$ exactly when $i + j \leq k + 1$), whose unique solution with boundary $(m_{i0}, m_{0j}) = (\delta_i, \delta_j)$ is the product law. The affine dimension is exactly k (exact homogeneous rank check, $k \leq 4$). Part 4 of Theorem 3.1: $B_0 \cdot y = 2m_{00} + m_{00} + 2m_{01} - m_{11} = 2 + 1 + 2\delta_1 - (2\delta_1 - 1) = 4$. This completes the proof of Theorem 3.1. $\square$

3.2 L2: the kernel law

Theorem 3.7 (kernel law). *Let $\delta^* = (\frac{1}{2}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2})$ (truncated to length k). For every $k \geq 2$:*

1. $\ker \Gamma_k(y_0(\delta^*)) = \text{span } N_k$ exactly (dimension $2k^2$);
2. $\text{rank } \Gamma_k = 2k + 1$, splitting as $(k+1, k)$ into party-symmetric and antisymmetric parts;
3. the same holds on an explicit open family of δ 's.

Proof. Reduction to the core Gram. The $2k + 1$ residual words $\{\psi, a_\ell \psi, b_\ell \psi : \ell = 1..k\}$ (a_ℓ = the unique alternating A -word of length ℓ ending in 1) together with N_k span $\mathbb{R}^{|\mathcal{W}_k|}$: every basis column reduces to residuals modulo relations (strip trailing 0's; if both parts end in 1, write $(a, b) = W \cdot (A_1, B_1)$ and use the dressing D_W , $|W| = |a| + |b| - 2 \leq k - 2$). Since $\Gamma(y_0(\delta))N_k = 0$ identically (Lemma 3.6), the form descends: $\text{rank } \Gamma_k = \text{rank } G_k(\delta)$, where G_k is Γ restricted to residuals, with closed-form entries ($\delta_0 := 1$)

$$[\psi, \psi] = 1, \quad [\psi, a_\ell] = \delta_{\lceil \ell/2 \rceil}, \quad [a_\ell, a_m] = \begin{cases} \delta_{\lceil \ell-m \rceil/2} & \ell \equiv m \pmod{2} \\ \delta_{(\ell+m+1)/2} & \ell \not\equiv m \pmod{2} \end{cases}, \quad [a_\ell, b_m] = \delta_{\lceil \ell/2 \rceil} + \delta_{\lceil m/2 \rceil} - 1.$$

So Theorem 3.7 is equivalent to: $G_k(\delta) \succ 0$.

The defect realization. On a commuting-party Hilbert space $H_A \otimes H_B$ take the mixed state

$$\rho = (1 - 2p) |x \otimes y\rangle \langle x \otimes y| + p \rho_A \otimes |y\rangle \langle y| + p |x\rangle \langle x| \otimes \rho_B,$$

where x, y are trivial vectors ($A_0 x = A_1 x = x$, $B_0 y = B_1 y = y$) and ρ_A is A_0 -invariant with dihedral spectral decomposition: two-dimensional blocks at reflection angle θ with A_0 -fixed vector e_θ , mixing measure μ , so that $c_j := \langle \alpha_j \rangle_{\rho_A} = \int \cos(j\varphi) d\mu(\varphi)$ ($\varphi = 2\theta$); ρ_B mirrors ρ_A . Then $A_0 \rho = B_0 \rho = \rho$, $(1 - A_1)(1 - B_1)\rho = 0$, and

$$\delta_j = 1 - p(1 - c_j), \quad m_{ij} = (1 - 2p) + pc_i + pc_j = \delta_i + \delta_j - 1 :$$

the product law holds automatically, so ρ realizes the level- k face point $y_0(\delta)$ for *every* k simultaneously.

The base point. The choice $p = \frac{1}{2}$, $d\mu = \frac{1}{\pi} \sin^2 \varphi d\varphi$ gives $c_j = -\frac{1}{2} [j = 2]$ (Iverson bracket, to avoid a clash with the face coordinates δ_j), hence $\delta = (\frac{1}{2}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \dots) = \delta^*$ exactly. Positive definiteness of $G_k(\delta^*)$: in the GNS space of ρ the residual vectors have components (on the ρ_A branch) $\psi \mapsto (1, 0)$, $a_{2j-1}\psi \mapsto (\cos j\varphi, \sin j\varphi)$, $a_{2j}\psi \mapsto (\cos j\varphi, -\sin j\varphi)$ in $L^2(\mu) \otimes \mathbb{C}^2$, and similarly for b_ℓ on the ρ_B branch. Since μ has a.e.-positive density, the trigonometric system $\{1, \cos j\varphi, \sin j\varphi\}$ is linearly independent in $L^2(\mu)$, and a short three-branch argument (any vanishing combination must vanish on each branch; the ρ_A branch kills the a -coefficients, the ρ_B branch the b 's, then the constant) gives independence of all $2k+1$ residuals. Hence $G_k(\delta^*) \succ 0$ for all k . (Machine checks: exact PD at $k \leq 5$ via tables and $k \leq 8$ via the closed-form entries; exact $(k+1, k)$ symmetric/antisymmetric splits.) A general interior family: any $p \in (0, 1)$ and any μ with at least $k+1$ support points (rational members via Chebyshev atoms). $\square$

Remark 3.8. *The base points used level-by-level in the certified ladder of [3] — $(\frac{1}{2}, \frac{1}{4})$, $(\frac{1}{2}, \frac{1}{4}, \frac{1}{2})$, $(\frac{1}{2}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2})$ — are truncations of this one canonical realization: the base family has a physical definition independent of the level.*

3.3 L4: the uniform rational Slater point

The second-order tangent program at the face (set up in detail in Section 5) constrains a second-order moment direction y_2 through the compressed form $N_k^\top \Gamma(y_2) N_k$ on the relation lattice. Its Slater point has a closed form.

Theorem 3.9 (uniform Slater law). *For every k , the rational point*

$$y_2^* = e_{\text{class}(1)} - y_0(\delta = 0)$$

(identity-class indicator minus the face point at $\delta = 0$; an affine — not PSD — point of the face parametrization) satisfies:

1. *the pins hold exactly: $(y_2^*)_1 = 1 - 1 = 0$ and $y_2^* = 0$ at the k face-coordinate classes;*
2. *$N_k^\top \Gamma(y_2^*) N_k = N_k^\top N_k \succ 0$, since $\Gamma(e_{\text{class}(1)}) = I$ (here $\text{rev}(p)q$ reduces to the empty word iff $p = q$) and $\Gamma(y_0(\delta)) N_k = 0$ identically (Lemma 3.6); positive definiteness on the whole lattice holds because the $2k^2$ relation vectors are independent (Proposition 2.1), and a fortiori on the program's blocks.*

Lemma 3.10 (blend lemma). *Let (τ, y_2) be a near-feasible rational jet for a block of the second-order tangent program (set up in Section 5; τ denotes the first-order jet datum — not the scalar transfer ratio t_0 of Sections 5–6) with exact pins and Schur complement $\text{Schur}(\tau, y_2) \succeq -\varepsilon I$ for a rational $\varepsilon \geq 0$. Let N be the block column of N_k , so that the compressed block of y_2^* is $N^\top N \succ 0$ (item 2 above). If $\varepsilon_p \in (0, 1)$ is a rational weight satisfying the explicit inequality*

$$\varepsilon_p \lambda_{\min}(N^\top N) > (1 - \varepsilon_p) \varepsilon,$$

then the blend $((1 - \varepsilon_p)\tau, (1 - \varepsilon_p)y_2 + \varepsilon_p y_2^)$ is exactly feasible: λ scales by $(1 - \varepsilon_p)$ (stays ≥ 0), pins stay exact (they are homogeneous, Lemma 5.3), and since the Schur subtrahend*

$Q(\tau) = F(\tau)^\top M_0^+ F(\tau)$ (notation of Theorem 5.4: F is linear in the first-order jet, M_0 the base block) is PSD-valued and quadratic, $Q((1 - \varepsilon_p)\tau) = (1 - \varepsilon_p)^2 Q(\tau) \preceq (1 - \varepsilon_p)Q(\tau)$, so

$$\text{Schur}(\text{blend}) \succeq (1 - \varepsilon_p) \text{Schur}(\tau, y_2) + \varepsilon_p N^\top N \succeq (\varepsilon_p \lambda_{\min}(N^\top N) - (1 - \varepsilon_p)\varepsilon)I \succ 0.$$

(The certified objective values are evaluated at the blended jet, so no separate tracking of the objective loss is needed.)

This removes, uniformly in k , the per-level question “does a rational Slater point exist”; it is also why the same certificate recipe succeeded at levels 2, 3, 4 in [3].

4 The witness

4.1 First-order structure: the gain direction u_k

Let y_1 be a first-order direction at the face (a class vector; the identity coordinate and face pins are imposed later, harmlessly, by Lemma 5.3). Define the compressed first-order form

$$P(y_1) := N_k^\top \Gamma(y_1) N_k.$$

We seek directions y_1 for which $P(y_1)$ is PSD of rank one on the lattice: rank-one alignment is exactly what the no-repair lemma consumes as its hypothesis (Lemma 6.1(b) below), and the $\mu = 0$ discussion of Remark 6.2 shows why unaligned first-order leakage is the regime where naive second-order repair breaks down. (We do not claim rank one is *necessary* for quadratic gain; it is the structure our certificates realize.) The rank-one direction is level-uniform and has a closed form.

Definition 4.1 (the gain vector u_k). *In the canonical kernel basis [trivial reductions in basis order | dressings in (ℓ_a, ℓ_b) order]:*

$$u_k[T_{(a,b)}] = \begin{cases} s(a)s(b) & \text{if exactly one of } a, b \text{ ends in } 0, \\ 0 & \text{otherwise,} \end{cases}$$

$$s(w) = \begin{cases} -1 & w \text{ starts with } 1, \\ +1 & \text{else (including } w = \emptyset); \end{cases}$$

$$u_k[D_W] = 2(-1)^{|W|} \quad (W \neq \emptyset), \quad u_k[D_\emptyset] = 0.$$

The formula is level-independent (old kernel vectors keep their words), so the tower property “ u_k extends u_{k-1} ” is automatic. At $k = 2$ it reproduces the certified jet of [3] exactly ($P(y_1^{\text{cert}}) = \frac{1}{4}u_2u_2^\top$); the linear system $P(y_1) = \lambda u_k u_k^\top$ was verified solvable with $\lambda \neq 0$ at $k = 2, \dots, 5$ before the closed-form witness below settled every k at once. Let

$$V_k = \{(y_1, \lambda) : P(y_1) = \lambda u_k u_k^\top, (y_1)_1 = 0\}$$

denote the first-order constraint space. No separate antisymmetric-block condition is needed: u_k is invariant under the party swap, so the antisymmetric compression of $\lambda u_k u_k^\top$ vanishes automatically for every element of V_k — this is Lemma 5.1(iii) below, where the swap-split notation N_s, N_a is also fixed.

Lemma 4.2 (identities on V_k). *For all k and all $(y_1, \lambda) \in V_k$, using only pairings of kernel vectors present at every level:*

$$(y_1)_{\langle A_0 \rangle} = -\frac{\lambda}{2}, \quad (y_1)_{\langle A_0 B_0 \rangle} = 0, \quad (y_1)_{\langle A_1 B_1 \rangle} = 2(y_1)_{\langle B_1 \rangle}, \quad B_0 \cdot y_1 = 0.$$

Proof. Evaluate $P(y_1) = \lambda u u^\top$ at explicit kernel pairs. (T_{A_0}, T_{A_0}) : $2y_1 - 2y_{\langle A_0 \rangle} = \lambda u [T_{A_0}]^2 = \lambda$, and $(y_1)_1 = 0$, so $y_{\langle A_0 \rangle} = -\lambda/2$. $(T_{A_0 B_0}, T_{A_0 B_0})$: $-2y_{\langle A_0 B_0 \rangle} = \lambda \cdot 0$ (both parts of (A_0, B_0) end in 0, so $u = 0$). $(T_{A_0}, T_{(A_0, B_1)})$: $2y_{\langle B_1 \rangle} - 2y_{\langle A_0 B_1 \rangle} = -\lambda$. $(D_\emptyset, D_\emptyset)$: $-8y_{\langle B_1 \rangle} + 4y_{\langle A_1 B_1 \rangle} = \lambda \cdot 0$. Summing: $B_0 \cdot y_1 = 2y_{\langle A_0 \rangle} + y_{\langle A_0 B_0 \rangle} + 2y_{\langle A_0 B_1 \rangle} - y_{\langle A_1 B_1 \rangle} = -\lambda + 0 + (2y_{\langle B_1 \rangle} + \lambda) - 2y_{\langle B_1 \rangle} = 0$. $\square$

Remark 4.3 (normalizations). *Two equivalent scalings occur. The canonical scaling uses u_k of Definition 4.1, in which $(y_1)_{\langle A_0 \rangle} = -\lambda/2$. The certificate scaling of the verification suite and of [3] uses $\tilde{u}_k = u_k/2$, hence $\tilde{\lambda} = 4\lambda$ and $(y_1)_{\langle A_0 \rangle} = -\tilde{\lambda}/8$. Sections 5 and 6 are stated in the certificate scaling (all constants there, e.g. $1/(1024t_0)$, are in that scaling); the independent re-verification confirmed the conversion numerically ($\tilde{\lambda} = 4$ measured for the canonical witness y^*).*

4.2 The closed-form witness y^*

For an alternating word w over $\{0, 1\}$ (possibly empty; 0, 1 standing for the letters X_0, X_1 of its party) define

$$i(w) = \#\{1\text{'s in } w\}, \quad \tau(w) = \text{first}(w) + \text{last}(w) - 1 \in \{-1, 0, 1\},$$

$$\sigma(w) = \text{first}(w) - \text{last}(w), \quad d_0(w) = [w = (0)],$$

with $\tau(\emptyset) = \sigma(\emptyset) = 0$. Every moment class at every level is a pair (α, β) of alternating words modulo dagger (reverse both) and party swap. Define, with $i = i(\alpha)$, $j = i(\beta)$, $\tau_a = \tau(\alpha)$ etc.:

$$y^*(\alpha, \beta) = i^2 j^2 - 2ij(\tau_a \tau_b + \sigma_a \sigma_b) + \frac{\tau_a + \tau_b}{2} - \frac{\tau_a^2 + \tau_b^2}{2} + \tau_a^2 \tau_b^2 + \frac{d_0(\alpha) + d_0(\beta)}{2}. \quad (2)$$

This is well defined on classes: dagger fixes i, τ and flips both σ 's (the product is invariant); party swap is manifest. Away from the degenerate words $(\emptyset)$ and (0) it collapses to the signed-square form

$$y^* = (ij + \varepsilon)^2 - c, \quad \varepsilon = -(\tau_a \tau_b + \sigma_a \sigma_b), \quad c = 1 - \left\lceil \frac{\tau_a + \tau_b}{2} \right\rceil,$$

exactly the shape mandated by the no-go of Section 8: a *signed* quadratic, not a Gram functional (y^* vanishes at the identity class but is nonzero elsewhere and takes both signs — it is not a state). It was discovered from the exact level- ≤ 5 solutions in a level-uniform gauge in which they stabilize level-to-level; the gauge-hunting record is part of the ancillary material.

Theorem 4.4 (the witness works at every level). *For every $k \geq 2$, with the canonical kernel basis $N_k = [T\text{'s} \mid D\text{'s}]$ and the closed-form u_k of Definition 4.1,*

$$N_k^\top \Gamma(y^*) N_k = u_k u_k^\top \quad \text{exactly,}$$

and y^ already satisfies the sharp pins: $y^*[\text{identity}] = 0$ and $y^* = 0$ at the k face-coordinate classes $(\emptyset, \beta_j)$ (there $i(\alpha) = 0$, $\tau_a = 0$, $\tau_b = 1$, so $y^* = \frac{1}{2} - \frac{1}{2} = 0$).*

Consistency checks: $y_{\langle A_0 \rangle}^* = y^*(\emptyset, (0)) = -\frac{1}{2} - \frac{1}{2} + \frac{1}{2} = -\frac{1}{2} = -\lambda/2$ at $\lambda = 1$ (the three nonzero terms of (2) in order), and $B_0 \cdot y^* = 0$, as Lemma 4.2 requires.

4.3 Proof of Theorem 4.4

Write each basis word as a pair of side words; the moment word of a pairing of basis words p, q is $m(p, q) = (m_A(a_p, a_q), m_B(b_p, b_q))$ with $m_A(v, w) = \text{reduce}(\text{rev}(v) \cdot w)$. For a kernel pair (K_1, K_2) with coefficient vectors c, c' the claimed equation is

$$E(K_1, K_2) := \sum_{p, q} c_p c'_q y^*(m(p, q)) - u(K_1) u(K_2) = 0.$$

The proof shows that E is, within each of finitely many combinatorial regimes, a polynomial of degree ≤ 2 in each of ≤ 4 free length coordinates, and that a finite exact-integer computation covers a full interpolation grid of every regime. *The finite computation is a constituent part of the proof* (in the tradition of computer-assisted proofs): the lemmas supply the degree and coverage bounds that make the finite check conclusive, and the check itself is exact integer arithmetic (no floating point anywhere).

Lemma 4.5 (nesting and locality). *(i) Two alternating words with the same first letter are nested (an alternating word is determined by its first letter and its length). (ii) Hence if $v[0] \neq w[0]$ (both nonempty), $\text{rev}(v) \cdot w$ is already reduced: $m_A = \text{rev}(v) \cdot w$ with $i(m) = i(v) + i(w)$, $\text{first}(m) = \text{last}(v)$, $\text{last}(m) = \text{last}(w)$; if v is the shorter of two same-first-letter words, $m_A = w[|v|:]$ (and symmetrically), with $i(m) = i(v) + i(w) - 2i(\text{shorter})$. (iii) Consequently, for every side pair: $\tau(m)$, $\sigma(m)$, $d_0(m)$ and $[m = \emptyset]$ are determined by the first letters, the length parities, and the values of $\min(|v|, |w|)$ and $|v| - |w|$ clamped at 2; and $i(m)$ is affine in $i(v), i(w)$ with coefficients determined by the same data.*

Proof. Cancellation in $\text{rev}(v) \cdot w$ runs exactly along the common prefix, which by (i) is the shorter word or empty; the boundary letters of the leftover tails sit at parity-determined positions. $\square$

Lemma 4.6 (tensor form of the kernel). *With $D^0(a) := e_a - e_{a'}$ (a ends in 0, a' its strip) and $D^\Delta(w) := e_{w'} - e_{w.1}$ (w ends in 0 or empty; w' = strip, $w.1$ = append 1):*

$$T_{(a,b)} = \begin{cases} D^0(a) \otimes e_b, & \text{only } a \text{ ends in } 0, u = s(a)s(b), \\ e_a \otimes D^0(b), & \text{only } b \text{ ends in } 0, u = s(a)s(b), \\ D^0(a) \otimes e_b + e_{a'} \otimes D^0(b), & \text{both end in } 0, u = 0, \end{cases}$$

$$D_W = D^\Delta(w_a) \otimes D^\Delta(w_b), \quad u = 2\omega(w_a)\omega(w_b), \quad \omega(w) = (-1)^{|w|},$$

with $u = 0$ for $W = \emptyset$. In particular u factorizes per party into regime constants.

Proof. $e_a \otimes e_b - e_{a'} \otimes e_{b'}$ telescopes; the rest is the definition of the dressing corners. $\square$

Lemma 4.7 (separability of y^*). *y^* is a finite sum $\sum_r f_r \otimes g_r$ of products of the seven per-word features $\{1, i^2, i\tau, i\sigma, \tau, \tau^2, d_0\}$ — immediate from (2).*

Lemma 4.8 (fine regimes; degree bound). *Call a fine regime of a pair of kernel elements the data: their types (T with $a/b/\text{both}$ trailing 0, or D), the first letters and length parities of the four base side words, and, per party, the values of $\min(\ell_1, \ell_2)$ and $\ell_1 - \ell_2$ clamped at 4 (exact if within the clamp, else “large”/sign only). Within a fine regime the admissible length tuples form a lattice with at most 4 free coordinates (per party: a min-coordinate and a difference-coordinate, each stepping by 2, present only in the “large” branches), and $E(K_1, K_2)$ is a polynomial of degree ≤ 2 in each free coordinate.*

Proof. The support words of a kernel element differ from its base words by at most one letter at the right end (strip trailing 0 / append 1), so each of the ≤ 16 pairings has side lengths $(\ell_1 + s, \ell_2 + s')$, $s, s' \in \{-1, 0, 1\}$; the element-level clamp at 4 makes every Lemma 4.5(iii) branch datum of every pairing constant across the regime, and validity of the elements (trailing-0 pattern) is parity-fixed. So, per pairing, each per-party feature of the moment word is a regime constant or affine in the free coordinates of that party (i of a word is affine in its length at fixed first letter and parity, slope $\frac{1}{2}$). By Lemma 4.7, $y^*(m(p, q))$ is a sum of (A-factor) $\times$ (B-factor) terms with each factor of degree ≤ 2 in its own party's coordinates (i^2 is the square of an affine function; $i\tau, i\sigma$ are affine times regime constants) — crucially, the A - and B -coordinates never multiply into one variable, so the degree per variable stays ≤ 2 . Finally $u(K_1)u(K_2)$ is a regime constant (Lemma 4.6). $\square$

Lemma 4.9 (grid coverage at side length 20). *Every fine regime realizes the full $\{0, 1, 2\}$ -grid of its free coordinates among element pairs with all side lengths ≤ 20 : the clamp threshold gives minimal “large” values ≤ 6 ($4 + \text{parity}$), and two grid steps add 4 to each of the min- and difference-coordinates: $6 + 4 + 6 + 4 = 20$. A polynomial of degree ≤ 2 in each variable vanishing on a $\{0, 1, 2\}$ -grid vanishes identically.*

Completion of the proof of Theorem 4.4. The exact-integer verifier `verify_r1_induction.py` evaluates $E(K_1, K_2)$ for all 1,413,721 kernel-element pairs with side lengths ≤ 20 (a level-independent enumeration) and finds $E = 0$ in every case; a margin run at ≤ 24 covers 2,883,601 pairs, and a rerun of the same engine at ≤ 26 (reproducible via `check_all_pairs(26)` in that file) covers 3,946,645 pairs. The independent re-verification added a *freshly written* implementation with 18,915 targeted extreme pairs plus 150,000 random pairs at side lengths up to 61 — all zero. By Lemmas 4.8 and 4.9 the ≤ 20 enumeration covers a full interpolation grid of every fine regime, so $E \equiv 0$ identically: for every kernel pair at every level. The verifier also checks $P(y^*) = u_k u_k^\top$ per level directly at $k = 2, \dots, 8$ and the sharp pins. (*The runs at ≤ 24 , ≤ 26 and the random sweep provide margin only; the ≤ 20 grid together with the degree bound constitutes the proof.*) $\square$

In addition, the regime step has been verified *fully symbolically*, removing the interpolation argument from the trust base: the verifier `verify_witness_symbolic.py` enumerates all 58,081 fine regimes from the definitions (clamp 3), assembles $E(K_1, K_2)$ per regime as an exact polynomial in the ≤ 4 free length coordinates (sympy, guarded branch decisions, none symbol-dependent), and finds E *identically zero* in every regime; the regime classifier is total, and the ≤ 20 enumeration realizes exactly the 58,081 regimes. Lemmas 4.8 and 4.9 thus remain as an independent second proof.

Remark 4.10 (the germ picture). *An exact induction-step analysis (steps $2 \rightarrow 3, 3 \rightarrow 4, 4 \rightarrow 5$; `explore_r1_induction.py`) shows that the old-class rows of the level- $(k+1)$ system reduce, modulo the level- k system, to exactly two residual conditions supported on the level- k “unseen” classes ($m_k =$ the alternating word $1(01)^{k-1}$ of length $2k - 1$):*

$$y[(\emptyset, m_k)] - y[((0), m_k)] = -\frac{1}{2}, \quad y[(\emptyset, m_k)] - y[((1), m_k)] - y[((0), 0m')] + y[((1), 0m')] = \frac{4k-3}{2},$$

($0m'$ the length- $(2k-1)$ word $0101 \dots 0$). Since unseen coordinates are free at level k , extendability is never obstructed — this is the mechanical content of the germ stabilization law (each extension cuts exactly two gauge dimensions, higher levels cut nothing further, and the bonding maps of the extendable-germ tower are surjective). The closed-form y^* satisfies both conditions identically and is the coherent infinite tower whose existence that law predicted.

5 Small- λ positivity: a feasible tangent jet at every level

5.1 The tangent program

From here on we use the certificate scaling of Remark 4.3 ($\tilde{u} = u_k/2$, $\lambda := \tilde{\lambda}$, so $(y_1)_{\langle A_0 \rangle} = -\lambda/8$ on V_k); all constants below are in this scaling.

Since the base point and all directions are swap-symmetric (class vectors), $[\Gamma(y), S] = 0$ for the party-swap permutation S of the basis words, $(v, w) \mapsto (w, v)$ ($S^2 = 1$), and the orthogonal congruence by the basis $[S\text{-fixed vectors and } e_x + e_{Sx} \mid e_x - e_{Sx}]$ block-diagonalizes every $\Gamma(y)$ into a symmetric block $M_s(y)$ (on $\ker(S - 1)$) and an antisymmetric block $M_a(y)$ (on $\ker(S + 1)$). Write M_{s0}, M_{a0} for the blocks at the base $y_0(\delta^*)$. By Theorem 3.7 the relation lattice $\Lambda_k := \text{range } N_k = \ker \Gamma_k(y_0(\delta^*))$ is exact; it is S -invariant and splits as $\Lambda_k = \Lambda_s \oplus \Lambda_a$, $\Lambda_s = \Lambda_k \cap \ker(S - 1)$, $\Lambda_a = \Lambda_k \cap \ker(S + 1)$ (Lemma 5.1(i) below). Define N_s (resp. N_a) as a matrix whose columns are the symmetric-block (resp. antisymmetric-block) coordinates of a basis of Λ_s (resp. Λ_a); the certificate files fix concrete integer choices. Then

$$M_{s0}, M_{a0} \succeq 0, \quad \ker M_{s0} = \text{span } N_s, \quad \ker M_{a0} = \text{span } N_a \quad \text{exactly:}$$

PSD-ness passes through the congruence, and for a PSD block, $M_{s0}v = 0 \iff v^\top M_{s0}v = 0 \iff \Gamma_0 Tv = 0 \iff Tv \in \Lambda_k \cap \ker(S - 1) = \Lambda_s$, with T the symmetric half of the congruence basis (likewise antisymmetric). The following lemma carries, for every k at once, the full-lattice witness identity of Theorem 4.4 to the swap-split hypotheses used below; it also discharges the antisymmetric-block clause dropped from the definition of V_k in Section 4.1.

Lemma 5.1 (swap-split bridge). *Fix $k \geq 2$. Then:*

1. *S permutes the canonical kernel basis: as coefficient vectors, $ST_{(a,b)} = T_{(b,a)}$ and $SD_{(w_a, w_b)} = D_{(w_b, w_a)}$. Writing π for the induced involutive permutation of the basis and Π for its matrix, $SN_k = N_k\Pi$; in particular Λ_k is S -invariant and splits as $\Lambda_s \oplus \Lambda_a$ with explicit bases $\{N_k e_m : \pi(m) = m\} \cup \{N_k(e_m + e_{\pi(m)}) : m < \pi(m)\}$ of Λ_s and $\{N_k(e_m - e_{\pi(m)}) : m < \pi(m)\}$ of Λ_a .*
2. *u_k is swap-invariant: $\Pi u_k = u_k$.*
3. *Let y be any class vector with $N_k^\top \Gamma(y) N_k = \lambda u_k u_k^\top$. Then the quadratic form $\Gamma(y)$ vanishes identically on Λ_a , has no Λ_s - Λ_a cross terms, and restricts to Λ_s as the rank-one form $N_k c \mapsto \lambda (u_k^\top c)^2$. Equivalently, in the block bases:*

$$N_s^\top M_s(y) N_s = \lambda \tilde{u} \tilde{u}^\top, \quad N_a^\top M_a(y) N_a = 0,$$

where $\tilde{u}_i = u_k^\top c^{(i)}$ for the lattice basis $\{N_k c^{(i)}\}$ underlying N_s ; and $\tilde{u} \neq 0$, since $u_k^\top (e_m + e_{\pi(m)}) = 2 u_k[m] \neq 0$ for any m with $u_k[m] \neq 0$.

Proof. (i) S exchanges the two tensor factors of Lemma 4.6, so it maps the support of $T_{(a,b)}$, namely $\{+(a, b), -(\text{strip } a, \text{strip } b)\}$, to $\{+(b, a), -(\text{strip } b, \text{strip } a)\}$, which is exactly $T_{(b,a)}$; and the four corners of $D_{(w_a, w_b)}$, coefficients included, to the four corners of $D_{(w_b, w_a)}$. (ii) Immediate from Definition 4.1: $u_k[T_{(a,b)}] = s(a)s(b)$ and $u_k[D_W] = 2(-1)^{|w_a|+|w_b|}$ are symmetric in the two parties. (iii) For $x = N_k c$ and $x' = N_k c'$,

$$x^\top \Gamma(y) x' = c^\top (N_k^\top \Gamma(y) N_k) c' = \lambda (u_k^\top c) (u_k^\top c').$$

If $\Pi c = -c$ then, by (ii), $u_k^\top c = (\Pi u_k)^\top c = u_k^\top \Pi c = -u_k^\top c$, so $u_k^\top c = 0$: the form vanishes whenever either argument lies in Λ_a (this covers both the antisymmetric block and the cross terms), and on Λ_s it is the stated rank-one square. The block identities follow since $N_s^\top M_s(y) N_s$ (resp. $N_a^\top M_a(y) N_a$) is precisely the matrix of the restriction of $\Gamma(y)$ to Λ_s (resp. Λ_a) in the chosen basis: the congruence preserves the pairing of lattice vectors. $\square$

All claims of Lemma 5.1 were additionally machine-verified in exact integer arithmetic at $k = 2, \dots, 5$ (`verify_bridge_lemma.py`: the permutation labels and involutivity, $\Pi u_k = u_k$, vanishing of the antisymmetric and cross compressions of $\Gamma(y^*)$, and the rank-one symmetric compression $\tilde{u}\tilde{u}^\top$), independently of the certificate bases; the certificate-basis instances at $k = 2, 3, 4$ are re-checked in the independent re-verification (audit S3).

The gain vector lives in the symmetric kernel; let K_s be a basis of $\tilde{u}^\perp$ there. The *second-order tangent program* at the face is: over pinned directions (y_1, y_2 vanishing at the identity class and the k face-coordinate classes) with $(y_1, \lambda) \in V_k$, $\lambda \geq 0$,

$$\text{maximize } B_0 \cdot y_2 - m \cdot y_1 \quad \text{s.t.} \quad \begin{pmatrix} M_{s0} & M_s(y_1) N_s K_s \\ * & (N_s K_s)^\top M_s(y_2) N_s K_s \end{pmatrix} \succeq 0, \quad (3)$$

plus the analogous antisymmetric block $\begin{pmatrix} M_{a0} & M_a(y_1) N_a \\ * & N_a^\top M_a(y_2) N_a \end{pmatrix} \succeq 0$. Its value v_k lower-bounds a_k once the framework step of Section 6 is in place. Any feasible point gives $v_k \geq$ its objective; by Lemma 4.2 the objective of a feasible point is $B_0 \cdot y_2 + \lambda/8$.

Lemma 5.2 (range identity). *On V_k , for every k , both block range conditions hold identically: $\text{range}(M_s(y_1) N_s K_s) \subseteq \text{range } M_{s0}$ and $\text{range}(M_a(y_1) N_a) \subseteq \text{range } M_{a0}$.*

Proof. M_{s0} is PSD with $\ker M_{s0} = \text{span } N_s$ exactly (Theorem 3.7), so $\text{range } M_{s0} = (\text{span } N_s)^\perp$ and the condition reads $N_s^\top M_s(y_1) N_s K_s = 0$. On V_k , $N_s^\top M_s(y_1) N_s = \lambda \tilde{u} \tilde{u}^\top$ and K_s is by definition a basis of $\tilde{u}^\perp$, so $N_s^\top M_s(y_1) N_s K_s = \lambda \tilde{u} (\tilde{u}^\top K_s) = 0$. The antisymmetric condition is $N_a^\top M_a(y_1) N_a = 0$, part of the definition of V_k . $\square$

Hence, by Albert's generalized Schur criterion, block-PSD is equivalent to Schur-complement-PSD at every point of V_k , for every k — the range identity observed per level in [3] is structural.

Lemma 5.3 (pin homogeneity and objective constants). *For every k :*

1. *All pins of the tangent program are homogeneous (y_1 and y_2 vanish at the identity class and the k face-coordinate classes), so V_k is a linear space and $t y_2$ is pin-feasible for all t if y_2 is.*
2. *The uniform Slater point y_2^* (Theorem 3.9) satisfies all pins, and $B_0 \cdot y_2^* = B_0 \cdot e_{\text{class}(1)} - B_0 \cdot y_0(0) = 0 - 4 = -4$ exactly (B_0 has no identity-class coefficient; $B_0 \cdot y \equiv 4$ on the affine face hull, including at $\delta = 0$, by Theorem 3.1(4)).*
3. *On V_k : $(y_1)_{\langle A_0 \rangle} = -\lambda/8$ and $B_0 \cdot y_1 = 0$ (Lemma 4.2), so the objective of (3) is $B_0 \cdot y_2 + \lambda/8$.*

5.2 The feasible point

Theorem 5.4 (small- λ positivity). *Fix $k \geq 2$ and suppose a witness exists at level k : a pinned direction $y_1(w)$ with*

$$N_s^\top M_s(y_1(w)) N_s = \tilde{u} \tilde{u}^\top, \quad N_a^\top M_a(y_1(w)) N_a = 0$$

(i.e. $(y_1(w), 1) \in V_k$). Let

$$C_s = (N_s K_s)^\top M_s(y_2^*) N_s K_s, \quad C_a = N_a^\top M_a(y_2^*) N_a \quad (\text{both } \succ 0 \text{ by Theorem 3.9}),$$

$$Q_s = F^\top M_{s0}^+ F, \quad F = M_s(y_1(w)) N_s K_s; \quad Q_a = G^\top M_{a0}^+ G, \quad G = M_a(y_1(w)) N_a,$$

and $\rho_k(w) = \min\{t : tC_s \succeq Q_s, tC_a \succeq Q_a\}$ (finite since $C_s, C_a \succ 0$; nonnegative since $Q_s, Q_a \succeq 0$). Then for every rational $t_0 > \rho_k(w)$,

$$v_k \geq \frac{1}{1024 t_0} > 0.$$

Proof. Take any $\lambda \in (0, \frac{1}{32t_0})$ and the explicit point

$$y_1 = \lambda y_1(w), \quad y_2 = (t_0 \lambda^2) y_2^*.$$

Pins hold by Lemma 5.3 (homogeneity); $\lambda \geq 0$; $(y_1, \lambda) \in V_k$ by linearity of V_k . Both block matrices in (3) are PSD by Albert's criterion: the corner blocks $M_{s0}, M_{a0} \succeq 0$ (Theorem 3.7), the range conditions hold (Lemma 5.2), and the Schur complements are

$$(t_0 \lambda^2) C_s - \lambda^2 Q_s = \lambda^2 (t_0 C_s - Q_s) \succeq 0,$$

and the same in the antisymmetric block. The objective is $B_0 \cdot y_2 + \lambda/8 = \lambda/8 - 4t_0 \lambda^2$ (Lemma 5.3), maximized over the allowed λ at $\lambda = \frac{1}{64t_0}$, with value $\frac{1}{1024t_0} > 0$. $\square$

By Theorem 4.4 the witness exists at every level: $w = y^*/4$ is pinned and $P(y^*/4) = \frac{1}{4} u_k u_k^\top$, so Lemma 5.1(iii) — applied with $\lambda = \frac{1}{4}$, i.e. $\tilde{u} = u_k/2$ in the certificate scaling — delivers exactly the two swap-split hypotheses $N_s^\top M_s(w) N_s = \tilde{u} \tilde{u}^\top$ and $N_a^\top M_a(w) N_a = 0$, for every $k \geq 2$ at once. (The same chain is verified exactly in the concrete certificate bases at $k = 2, 3, 4$ in the independent re-verification.) Hence $v_k > 0$ for every $k \geq 2$.

6 Exact arc feasibility: from tangent jet to $c_k(s)$

The step from a feasible tangent jet to the true SDP value is where naive second-order arguments fail: when the first-order compressed form degenerates on the kernel, the standard $o(s^2)$ -correction argument breaks down. Our jets degenerate on $\tilde{u}^\perp$ — but exactly there the second-order block is strictly positive definite, and the rank-one first-order form has exactly zero kernel cross-coupling. That structure makes any correction unnecessary: the arc is exactly feasible as written.

Lemma 6.1 (no-repair positivity). *Let $M(s) = M_0 + sM_1 + s^2M_2$ be symmetric $n \times n$ and decompose $\mathbb{R}^n = R \oplus U \oplus W$ orthogonally with $\dim U \leq 1$. Assume*

- (a) $M_0 \succeq 0$ with $\ker M_0 = U \oplus W$ exactly; $c := \min \text{spec}(M_0|_R) > 0$;
- (b) $P_{\ker} M_1 P_{\ker} = \hat{\lambda} P_U$ with $\hat{\lambda} > 0$ (rank one, aligned with U ; if $U = 0$, the kernel-compressed first-order form vanishes). In particular the U - W and W - W first-order blocks are exactly zero;
- (c) no condition on the range-kernel coupling $X_1 := P_R M_1 P_{\ker}$;

(d) $H_{WW} := P_W M_2 P_W - X_{1W}^\top M_0^+ X_{1W} \succeq \mu I_W$ with $\mu > 0$ ($X_{1W} = P_R M_1 P_W$).

Set $b_1 = \|M_1\|$, $b_2 = \|M_2\|$ (spectral norms), $h = \|H_{UU}\|$, $\kappa = \|H_{UW}\|$ (blocks of $H := P_{\ker} M_2 P_{\ker} - X_1^\top M_0^+ X_1$; note $h, \kappa \leq b_2 + b_1^2/c$), and

$$\varepsilon := \frac{2b_1^2(b_1 + b_2)}{c^2} + \frac{4b_1 b_2}{c} + \frac{2b_2^2}{c}, \quad s^* := \min \left\{ 1, \frac{c}{2(b_1 + b_2)}, \frac{\hat{\lambda}}{2(h + \varepsilon)}, \frac{\mu}{2\varepsilon}, \frac{\hat{\lambda}\mu}{4(\kappa + \varepsilon)^2} \right\}$$

(dropping the $\hat{\lambda}$ -terms if $U = 0$). Then $M(s) \succeq 0$ for all $s \in (0, s^*]$ (and $\succ 0$ on the range-block-kernel-complement in the interior). No $o(s^2)$ correction is needed.

Proof. For $s \leq \min\{1, \frac{c}{2(b_1 + b_2)}\}$: $\|A(s) - A_0\| \leq s b_1 + s^2 b_2 \leq s(b_1 + b_2) \leq c/2$, so the range block $A(s) := P_R M(s) P_R \succeq \frac{\varepsilon}{2} I \succ 0$ and $\|A(s)^{-1}\| \leq 2/c$. By the Schur criterion, $M(s) \succeq 0$ iff the kernel Schur complement $S(s) := K(s) - X(s)^\top A(s)^{-1} X(s) \succeq 0$, where $K(s) = s \hat{\lambda} P_U + s^2 G_2$ ($G_2 = P_{\ker} M_2 P_{\ker}$; the s -term off U vanishes *exactly* by (b), the decisive consequence of the rank-one form) and $X(s) = s X_1 + s^2 X_2$. Expanding around $A_0^{-1} = M_0^+|_R$:

$$X(s)^\top A(s)^{-1} X(s) = s^2 X_1^\top M_0^+ X_1 + \text{Rem}(s),$$

$$\|\text{Rem}(s)\| \leq s^2 b_1^2 \cdot \frac{2s(b_1 + b_2)}{c^2} + 2s^3 b_1 b_2 \cdot \frac{2}{c} + s^4 b_2^2 \cdot \frac{2}{c} \leq s^3 \varepsilon \quad (s \leq 1),$$

using $\|A(s)^{-1} - A_0^{-1}\| \leq \|A_0^{-1}\| \|A(s) - A_0\| \|A(s)^{-1}\| \leq 2s(b_1 + b_2)/c^2$. Hence $S(s) = s \hat{\lambda} P_U + s^2 H - \text{Rem}(s)$ with the block bounds

$$U\text{-entry } \alpha \geq s \hat{\lambda} - s^2 h - s^3 \varepsilon \geq \frac{s \hat{\lambda}}{2} \quad \text{if } s \leq \frac{\hat{\lambda}}{2(h + \varepsilon)};$$

$$W\text{-block } D \succeq s^2 \mu I_W - s^3 \varepsilon I_W \succeq \frac{s^2 \mu}{2} I_W \quad \text{if } s \leq \frac{\mu}{2\varepsilon}; \quad \text{cross } \|r\| \leq s^2 \kappa + s^3 \varepsilon \leq s^2(\kappa + \varepsilon).$$

For the 2×2 block form $\begin{pmatrix} \alpha & r^\top \\ r & D \end{pmatrix}$ with $D \succ 0$, PSD holds iff $\alpha \geq r^\top D^{-1} r$, for which $\alpha \cdot \lambda_{\min}(D) \geq \|r\|^2$ suffices: $\frac{s \hat{\lambda}}{2} \cdot \frac{s^2 \mu}{2} \geq s^4(\kappa + \varepsilon)^2$ iff $s \leq \hat{\lambda}\mu/(4(\kappa + \varepsilon)^2)$. If $U = 0$ then $S(s) \succeq s^2(\mu - s\varepsilon)I \succeq 0$ directly. $\square$

Remark 6.2 (necessity of the strict margin). If $\mu = 0$ — the W -margin merely PSD, as happens at exactly optimal jets by complementary slackness — the s^3 remainder can go negative on W and a genuine third-order repair is required; this is precisely the measured third-order infeasibility of the raw certified ladder jets of [3] (their blends are minute — blend weight $\varepsilon_p = 10^{-54}$ at $k = 2$, 10^{-7} at $k = 3, 4$ — so the repaired margins μ , and with them s^* , are minuscule, and violations are visible at $s \sim 0.05$). Hypothesis $\mu > 0$ is the sharp boundary: the framework caveat flagged in [3] was the $\mu = 0$ case, not a hole in the strict case.

Theorem 6.3 (exact arc; witness $\Rightarrow a_k > 0$). Fix $k \geq 2$ and a witness w as in Theorem 5.4. Let $(y_1, y_2) = (\lambda y_1(w), t_0 \lambda^2 y_2^*)$ with $t_0 > \rho_k(w)$ rational and $\lambda = \frac{1}{64t_0}$, and set

$$y(s) = y_0(\delta^*) + s y_1 + s^2 y_2.$$

Then there is an explicit $s_k^* > 0$ (Lemma 6.1's constants, computed per swap block) such that for all $s \in (0, s_k^*]$:

1. $y(s)_1 = 1$ and $\Gamma_k(y(s)) \succeq 0$: the arc is exactly feasible for the level- k SDP at deformation s ;

2. $B_s \cdot y(s) = 4 - s + g_k s^2 + (t_0 \lambda^2) s^3$ exactly (a cubic, not an expansion), with

$$g_k = \frac{\lambda}{8} - 4t_0 \lambda^2 = \frac{1}{1024 t_0} > 0.$$

Hence $c_k(s) \geq 4 - s + g_k s^2$ on $(0, s_k^*]$ (the cubic term is positive), so $a_k \geq g_k > 0$.

Proof. Swap-invariance of $y(s)$ gives $[\Gamma(y(s)), S] = 0$, so the congruence block-diagonalizes: $\Gamma \succeq 0 \iff M_s(s) \succeq 0$ and $M_a(s) \succeq 0$. Apply Lemma 6.1 to each block. *Symmetric block:* $R = \text{range } M_{s0}$, $\ker M_{s0} = \text{span } N_s$ exactly (Theorem 3.7); $W = \text{range}(N_s K_s)$, U its orthocomplement in the kernel (1-dimensional: the kernel quadratic form is $q_1(N_s v) = \lambda(\tilde{u}^\top v)^2$ and N_s is injective, so $\hat{\lambda} = \lambda(\tilde{u}^\top v_U)^2 > 0$). Hypothesis (b) holds because $N_s^\top M_s(y_1) N_s = \lambda \tilde{u} \tilde{u}^\top$ makes the kernel bilinear form $b_1(N_s v, N_s v') = \lambda(\tilde{u}^\top v)(\tilde{u}^\top v')$, which vanishes whenever either argument lies in W . Hypothesis (d) is the strict t_0 -margin: in $N_s K_s$ -coordinates

$$(N_s K_s)^\top M_s(y_2) N_s K_s - Q_s(y_1) = \lambda^2 (t_0 C_s - Q_s) \succeq \lambda^2 m I \quad (m = \lambda_{\min} > 0, \text{ certified exactly}),$$

and the orthonormal-frame congruence $E_W = (N_s K_s)^\top T$ rescales the margin by $\sigma_{\min}(T)^2 = 1/\|N_s K_s\|^2 > 0$. *Antisymmetric block:* the $U = 0$ case ($N_a^\top M_a(y_1) N_a = 0$ on V_k), margin $\lambda^2(t_0 C_a - Q_a) \succ 0$. Feasibility (1) follows on $(0, s_k^*]$ with s_k^* the minimum of the two blocks' s^* ; the pins give $y(s)_1 = 1 + 0 + 0$.

Objective (2): $B_s = B_0 - s m \cdot$, and

$$B_s \cdot y(s) = B_0 \cdot y_0 + s(B_0 \cdot y_1 - m \cdot y_0) + s^2(B_0 \cdot y_2 - m \cdot y_1) + s^3(-m \cdot y_2) = 4 - s + g_k s^2 + t_0 \lambda^2 s^3,$$

using $B_0 \cdot y_0 = 4$, $m \cdot y_0 = 1$, $B_0 \cdot y_1 = 0$, $m \cdot y_1 = -\lambda/8$, $B_0 \cdot y_2 = -4t_0 \lambda^2$, and $m \cdot y_2 = -t_0 \lambda^2$ (Lemma 5.3 together with $(y_2^*)_{\langle A_0 \rangle} = -1$). Feasibility gives $c_k(s) \geq B_s \cdot y(s)$, and the cubic term is positive. $\square$

Two machine checks accompany the lemma ($k = 2, 3, 4$). `verify_framework_lemma.py` re-verifies the structural hypotheses of Lemma 6.1 exactly (base kernels, rank-one alignment, strict t_0 -margins), verifies $M_s(s), M_a(s)$ exactly positive definite at $s = 10^{-3}, 10^{-4}$ by rational pivots with the objective cubic exact, and reports the lemma constants and s_k^* in floating point. `verify_sstar_rational.py` then certifies the constants themselves in exact rational arithmetic — lower bounds on c and μ by rational PD certificates against exact Gram matrices, $\hat{\lambda}$ and h as exact rationals (the U -direction is rational), upper bounds on b_1, b_2, κ by exact PD and square comparisons — and outputs certified rational interval endpoints $s_2^* \geq 1.6 \times 10^{-3}$, $s_3^* \geq 9.2 \times 10^{-3}$, $s_4^* \geq 7.5 \times 10^{-3}$ (the exact fractions are printed by the verifier), each $\geq 10^{-3}$, so both exact-PD sample points lie inside the certified intervals. The analysis is quantitatively tight, not just a bound: the measured minimal eigenvalue at $k = 2$, $s = 10^{-3}$ is $+2.74 \times 10^{-10}$, which matches $s^2 \mu$ of the symmetric block; the lemma's guaranteed floor is the halved, min-over-blocks value $s^2 \mu / 2 \approx 1.5 \times 10^{-11}$, comfortably below it.

Corollary 6.4 (the certified ladder bounds transfer to a_k). *Lemma 6.1 applies verbatim to any strictly feasible point of the tangent program (rank-one first-order form, exact base kernel, strictly PD Schur blocks): its objective value lower-bounds a_k . The certified ladder jets of [3] (`verify_v{2,3,4}_rational_base.py`) are blends with strictly positive-definite Schur complements (Lemma 3.10), rank-one first-order forms and the exact base kernel, so their certified values transfer unconditionally:*

$$a_2 > \frac{1}{39}, \quad a_3 > \frac{1}{188}, \quad a_4 > \frac{1}{641}.$$

The framework caveat of [3] is thereby discharged at $k = 2, 3, 4$ (the raw jets' s^* is tiny because the blend weights are $\varepsilon_p = 10^{-54}$ at $k = 2$ and 10^{-7} at $k = 3, 4$, hence so are the repaired Schur margins, which is why they look third-order infeasible at $s \sim 0.05$; the lemma needs only some interval $(0, s^*]$, and the $\liminf$ does the rest).

7 Proof of the Main Theorem

Proof of Theorem 1.1. Lower interval bound, $k \geq 2$. By Theorem 4.4 the witness $w = y^*/4$ exists at every level (pinned, certificate-scaled, antisymmetric compression zero). By Theorem 3.9, $C_s, C_a \succ 0$ for all k , so $\rho_k(w) < \infty$ and a rational $t_0 = t_0(k) > \rho_k(w)$ exists. Theorem 6.3 then gives explicit $s_k^* > 0$ and $g_k = 1/(1024 t_0(k)) > 0$ with $c_k(s) \geq 4 - s + g_k s^2$ on $(0, s_k^*]$, hence $a_k \geq g_k > 0$.

Non-exactness. By (1) there is $s_1 > 0$ with $c_Q(s) \leq 4 - s + \frac{s^3}{3}$ on $(0, s_1]$. For $\varepsilon_k := \min\{s_k^*, s_1, 2g_k\}$ and $s \in (0, \varepsilon_k]$,

$$c_k(s) - c_Q(s) \geq g_k s^2 - \frac{s^3}{3} \geq g_k s^2 \left(1 - \frac{2}{3}\right) > 0.$$

For the remaining finite levels: the level-1 and almost-quantum relaxations are outer to level 2 ($\mathcal{W}_1 \subset \mathcal{W}_{1+AB} \subset \mathcal{W}_2$, the latter inclusion of monomial sets verified explicitly in the independent re-verification), so $c_1(s) \geq c_{1+AB}(s) \geq c_2(s) > c_Q(s)$ on $(0, \varepsilon_2]$. Hence *every* finite level fails to be exact on an explicit interval $(0, \varepsilon_k]$.

The exponent. The lower bound $c_k(s) - (4 - s) \geq g_k s^2$ on $(0, s_k^*]$ is unconditional. For the matching upper bound we invoke the almost-quantum expansion $c_{1+AB}(s) = 4 - s + \frac{3}{64}s^2 + o(s^2)$ of [3]; its $\leq$ half rests on the dual second-order arc constructed there, whose tangent conditions sit at the boundary of the PSD cone (the $\mu = 0$ regime of Remark 6.2), so we state the upper half as conditional on that expansion. Monotonicity of the hierarchy then gives

$$g_k s^2 \leq c_k(s) - (4 - s) \leq c_2(s) - (4 - s) \leq c_{1+AB}(s) - (4 - s) = \frac{3}{64}s^2 + o(s^2),$$

the first inequality on $(0, s_k^*]$; so, granted the companion's expansion, the level- k overshoot is $\Theta(s^2)$ with $g_k \leq a_k \leq \frac{3}{64}$. (None of the non-exactness statements above depend on this step.)

Quantitative instances. Corollary 6.4 gives the unconditional $a_2 > \frac{1}{39}$, $a_3 > \frac{1}{188}$, $a_4 > \frac{1}{641}$. $\square$

Remark 7.1 (explicit constants). *The construction is fully effective. With the certified per-level witnesses of the ladder, $t_0(k) = 0.316, 29.7, 302$ at $k = 2, 3, 4$, giving $g_2 \geq \frac{15625}{5053008} \approx 3.09 \times 10^{-3}$, $g_3 \approx 3.29 \times 10^{-5}$, $g_4 \approx 3.23 \times 10^{-6}$; with the universal witness $y^*/4$ itself the independent re-verification gave better constants at higher levels, $t_0 = 0.348, 8.59, 69.4$. The interval endpoints at $k = 2, 3, 4$ are certified rational: $s_2^* \geq 1.6 \times 10^{-3}$, $s_3^* \geq 9.2 \times 10^{-3}$, $s_4^* \geq 7.5 \times 10^{-3}$ (`verify_sstar_rational.py`, exact arithmetic throughout; the floating-point evaluation of the lemma formula gives 1.8×10^{-3} , 1.0×10^{-2} , 8.4×10^{-3}). At a general level k , s_k^* may be taken to be any positive rational below the lemma's minimum; since all constants are eigenvalue/norm data of explicit rational matrices, the same rational-PD certification procedure applies verbatim at every k . The witness-dependent growth of $t_0(k)$ means the theorem provides positivity uniform in kind, not in size: no decay law for a_k is claimed (Section 9).*

Remark 7.2 (reading of a_k). a_k is defined as a $\liminf$; Theorem 6.3 in fact proves the stronger interval statement $c_k(s) \geq 4 - s + g_k s^2$ pointwise on $(0, s_k^*]$, so no expansion or analyticity assumption on c_k enters the non-exactness conclusion. Whenever c_k admits a second-order expansion at 0^+ , its coefficient is $\geq g_k$.

8 The mechanism: the overshoot is non-quantum

Why did the witness have to be the signed formula (2)? The following two no-gos locate the level- k overshoot strictly outside quantum theory's tangent directions; together they are the conceptual heart of the result.

Proposition 8.1 (GNS no-go: no state witness). *For $k \geq 3$, no witness y_1 with $P(y_1) = u_k u_k^\top$ can be realized as the moment functional $\langle \psi_1 | \cdot | \psi_1 \rangle$ of a vector in any Hilbert-space representation of the scenario algebra (commuting involutions A_i, B_j).*

Proof. If y_1 consists of moments of ψ_1 , then $P(y_1)$ is the Gram matrix of the kernel-defect vectors $K_a \psi_1$, and rank-one positivity in a definite space forces $K_a \psi_1 = u_a v$ with $|v| = 1$. The u_k -amplitudes then force: (1) $(A_0 - 1)\psi_1 = (B_0 - 1)\psi_1 = v$ (from the trailing-0 reductions with $|u| = 1$); (2) $A_i v = B_j v = -v$ (from the length-3 reductions, present once $k \geq 3$); (3) $R_1 \psi_1 = 0$ ($u[D_\emptyset] = 0$). Self-adjointness gives $\langle v, R_1 \psi_1 \rangle = \langle R_1 v, \psi_1 \rangle = 4\langle v, \psi_1 \rangle = 0$ and $\langle v, (A_0 - 1)\psi_1 \rangle = \langle (A_0 - 1)v, \psi_1 \rangle = -2\langle v, \psi_1 \rangle = 0$, i.e. $|v|^2 = \langle v, (A_0 - 1)\psi_1 \rangle = 0$: contradiction. $\square$

Remark 8.2 (indefinite (Krein) representations). *The proposition is stated for Hilbert-space states only, and deliberately so. In a Krein space the argument changes at its first step: an indefinite rank-one Gram $[K_a \psi_1, K_b \psi_1] = u_a u_b$ does not force the defect vectors to be collinear — each $K_a \psi_1 = u_a v + n_a$ may be perturbed by mutually orthogonal neutral vectors n_a ($[n_a, n_b] = 0$, $[v, n_a] = 0$) without changing the Gram — so the chain (1)–(3) need not start. What the same computation does show is conditional: if the defects are collinear, $K_a \psi_1 = u_a v$, then self-adjointness forces $[v, v] = 0$, i.e. the gain-carrying direction must be neutral, which is impossible in a definite space but not in an indefinite one. We therefore make no claim about general Krein-space realizations. (Nothing in the main theorem depends on this section; the propositions serve to locate the witness outside quantum tangent directions.)*

Proposition 8.3 (curve no-go: no smooth quantum family). *For any smooth curve of genuine commuting-party quantum models $(A(t), B(t), \psi(t))$ through the base realization, the compressed first-order form cannot equal $\lambda u_k u_k^\top$ with $\lambda > 0$ at any order in t , for any $k \geq 3$.*

Proof. The dressed defect vectors obey $\chi_{D_W}(t) = A_{w_a}(t) B_{w_b}(t) \chi_{D_\emptyset}(t)$ with unitary prefactors, so $\|\chi_{D_W}(t)\| = \|\chi_{D_\emptyset}(t)\|$ for all t : the compressed form has equal diagonal at all dressings at every order in t . But the gain direction requires $u[D_W]^2 = 4$ against $u[D_\emptyset]^2 = 0$. $\square$

Thus the direction that produces the quadratic overshoot is invisible not only to states (Proposition 8.1) but to all smooth families of quantum models (Proposition 8.3): *the overshoot lives strictly in the NPA-feasible, non-quantum part of the tangent cone at the critical face.* Finite NPA levels fail near the critical line because they admit a signed tangent direction that quantum mechanics forbids — and one level-independent signed quadratic (y^*) realizes it at every level simultaneously, in the coherent-tower form predicted by the germ stabilization law (Remark 4.10).

9 Discussion

9.1 What remains open

The decay law. The theorem gives $a_k \geq 1/(1024 t_0(k)) > 0$ with $t_0(k)$ witness-dependent and growing (0.35, 8.6, 69 along y^*); the measured coefficients $a_2 \approx 0.0281$, $a_3 \approx 0.0096$ [3] decay

with irregular successive ratios (≈ 0.60 , then ≈ 0.34). Nothing here controls the decay; a uniform-in- k lower bound on $\min_w \rho_k(w)^{-1}$, or a level-to-level inequality $a_{k+1} \geq r' a_k$, would sharpen positivity into a rate.

The dimension law (R2). The first-order solution space V_k has $\dim V_k = 2k + 3$ verified at $k \leq 5$, with a constructive lower bound for all k ; the matching upper bound is open. It is *not used* anywhere in the proof — the certificates never invoke it — but it is the one remaining structural conjecture of this framework.

The fixed- s question. Our theorem is local: for each k , non-exactness on $(0, \varepsilon_k]$ with $\varepsilon_k \downarrow 0$ allowed. Since each level *is* exact for s above its threshold, the pointwise required level $D(s)$ is finite for every fixed $s > 0$; what we prove is $D(s) \rightarrow \infty$ as $s \rightarrow 0^+$. The stronger claim that some fixed s admits no exact level is false in spirit here and we do not make it.

9.2 Placement

The result sharpens the picture drawn in [3] and closes its conjecture. In the degree-bounded noncommutative Fejér–Riesz theorem of Klep–Levenson–McCullough [6], strict positivity at margin ε yields a factorization at some finite one-party degree $M'(\varepsilon)$; our family has margin $\sim s^3$ and the theorem proves the complementary necessity statement — no bounded level (hence no bounded word length) certifies uniformly as $s \rightarrow 0$. In the finite-convergence theory of polynomial optimization, Nie’s theorem [7] (through Marshall’s boundary Hessian condition [8]) makes finite exactness *generic*; the doubly-tilted critical point is exactly a failure point of the boundary Hessian condition (cubic touch, contact order three), and our theorem shows that there the generic conclusion genuinely fails — the first unconditional instance of this failure mode for the NPA hierarchy at an attained, self-tested optimum, obtained by construction rather than by hardness reduction [10]. The commutative shadow of the problem cannot see this: on the compact torus the degenerating family is SOS at base degree ([3], following [11, 12]), so the divergence is carried by the noncommutative moment structure — in the present language, by the signed non-quantum tangent direction of Section 8. The scaling limit of the critical degeneracy is a rescaled Motzkin form [3], but the finite-level failure proved here is not a scalar-polynomial degree phenomenon; it is a property of the restricted certificate class of the hierarchy, localized now in an explicit direction. Finally, the single-tilt functional, whose touch is quadratic, is finitely exact at level $1+AB$ with the Bamps–Pironio certificate [9] — the dichotomy between quadratic and cubic contact is complete: contact order two is finitely certifiable, and at the contact-order-three point no finite level suffices.

A Verification programs

The complete verification suite, together with both papers, is available at <https://github.com/tohafrit/npa-nonexactness>; `run_all_checks.sh` reproduces every machine check below in about forty seconds. Every claim entering the proofs is either a finite hand-checkable argument or an exact machine computation (integer or rational arithmetic throughout; no floating point enters any proof). The finite computations are constituent parts of the proofs, with the degree/coverage lemmas (4.8, 4.9) making them conclusive. The ancillary files and what each one establishes:

file	checks	runtime
verify_pipeline_laws.py	counting laws; Lemma 3.2 ($E \succ 0$, affine identity); Lemma 3.4 (rank 15/17); product-law identity $\Gamma(y_0(\delta))N_k = 0$; affine dimension k ; Theorem 3.7 (exact PD of G_k , $k \leq 8$; $(k+1, k)$ splits); u_k facts and Lemma 4.2 on full solution spaces ($k \leq 5$); Theorem 3.9 ($K^\top \Gamma(y_2^*)K = K^\top K$, $k \leq 5$)	~ 2 s
verify_r1_induction.py	Theorem 4.4: per-level $P(y^*) = u_k u_k^\top$ at $k = 2..8$; sharp pins; the regime-grid engine, $E(K_1, K_2) = 0$ for all 1,413,721 pairs at side length ≤ 20 (margin: 2,883,601 at ≤ 24); exact integers	~ 4 s
verify_bridge_lemma.py	Lemma 5.1: swap permutation of the T/D basis, $\Pi u_k = u_k$, vanishing antisymmetric and cross compressions, rank-one symmetric compression $\tilde{u}\tilde{u}^\top$; exact integers, $k = 2..5$, independent of the certificate bases	< 1 s
verify_small_lambda.py	Theorem 5.4: range identity, pins, $C_s, C_a \succ 0$, ρ_k , the feasible point and its value, $k = 2, 3, 4$, exact	< 1 s
verify_framework_lemma.py	Lemma 6.1 and Theorem 6.3: structural hypotheses re-verified exactly; exact rational PD of $M_s(s)$, $M_a(s)$ at $s = 10^{-3}, 10^{-4}$; exact cubic objective; float report of the lemma constants; $k = 2, 3, 4$	< 1 s
verify_sstar_rational.py	certified rational bounds for every Lemma 6.1 constant (c, μ lower, b_1, b_2, κ upper, $\hat{\lambda}, h$ exact) and certified rational s_k^* ($\geq 1.6 \times 10^{-3}$, 9.2×10^{-3} , 7.5×10^{-3} at $k = 2, 3, 4$); exact rational arithmetic	< 1 s
verify_v{2,3,4}_rational_base.py	the certified ladder jets behind Corollary 6.4: exact rational tangent-program certificates for $v_2 > \frac{1}{39}$, $v_3 > \frac{1}{188}$, $v_4 > \frac{1}{641}$ (data in v{2,3,4}_cert_data.py)	seconds–minutes
audit_spot.py	the independent re-verification S1–S5 (per-item provenance below), including the composition step: y^* evaluated in the certificate bases at $k = 2, 3, 4$ gives exactly $N_s^\top M_s(y^*)N_s = 4uu^\top$, antisymmetric compression 0, pins, range identity	~ 11 s
scenario_ext.py, npa_general.py	shared library: level- k word/class/moment-matrix machinery used by the verifiers	—

Each verifier exits 0 iff all its checks pass; all pass. Supporting exploration scripts (attack_r1.py, explore_r1_induction.py, r1_formula_hunt.py) document, respectively, the dual criterion and germ analysis of Remark 4.10 and the gauge in which y^* was found; they play no role in the proofs.

Independent re-verification. The complete chain was checked a second time, directed at the points where an error would be most damaging. The machine checks of this audit are collected in audit_spot.py (sections S1–S5); the degree of independence varies by item, so we state it per item. *Freshly written code* (S1, S2: own word/reduction/kernel/ u_k/y^* implementations, sharing nothing with the development stack): a reproduction of $P(y^*) = u_k u_k^\top$ at $k = 2, \dots, 6$ with pins and anchors (S1), and a stress test of the fine-regime completeness of Lemmas 4.8–4.9 with 18,915 targeted extreme pairs and 150,000 random pairs at side lengths up to 61 (S2), backed by

a hand re-derivation of Lemma 4.5(iii). *Independent driver scripts over the shared exact stack* (S3, S4 import `scenario_ext` and the `LevelData` machinery of `verify_small_lambda.py`): the composition of Theorem 4.4 with Theorem 5.4 in the concrete certificate bases at $k = 2, 3, 4$ (the one step not covered by the original programs, now checked exactly) together with the normalization bookkeeping of Remark 4.3 (S3); and exact positive definiteness of the full *unsymmetrized* $\Gamma(y(s))$ at $s = 10^{-3}$, bypassing the swap splitting entirely — run at $k = 2$ and 3 only (the swap-split exact PD checks of `verify_framework_lemma.py` cover $k = 2, 3, 4$) — with the objective cubic recomputed from the raw functional (S4). The audit also re-derived Lemma 6.1 by hand. Two further items quoted above live elsewhere: the extended enumeration at side length ≤ 26 (3,946,645 pairs, all zero) is a rerun of the development engine, reproducible via `check_all_pairs(26)` in `verify_r1_induction.py`, and the base-point integrals of Theorem 3.7 are checked exactly in `verify_pipeline_laws.py` (section D). No discrepancy was found in any step.

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